

Title: What Should We Play Next? A Board Game Recommender for Groups Using BoardGameGeek Ratings

Dataset

BoardGameGeek reviews (Kaggle: [jvanelteren/boardgamegeek-reviews](#)), on Jojie at [/mnt/data/public/bgg/](#). Ratings from `bgg-19m-reviews.csv` (user, ID, rating; comments unused): 18,964,794 ratings after dropping 13 below 1, from 412,815 users across 21,839 games, with no duplicate user-game pairs. After keeping users with at least 20 ratings: 169,027 users and 17,613,867 ratings. Player counts and other attributes come from `games_detailed_info.csv` (21,631 games). 213 rated games have no metadata, and 173 games (0.8%) have an unknown minimum or maximum player count and are excluded from player-filtered results.

Problem

A group of friends often can't decide which board game to try next, and browsing 20,000+ titles doesn't help. Given the games a group has played, which ones they liked and disliked, and how many players they are, can we recommend games they haven't played that they're likely to enjoy? A group is treated as one user who pools their ratings.

Solution abstract

Keep users with at least 20 ratings and split into training and held-out users. Train ALS on training users, comparing it against popularity and item-item CF. For each held-out user, split their ratings into an “input” set (liked and disliked games) and a held-out set. Fold the user into the ALS game-factor space using only the liked games as positive signal; apply a similarity-based penalty from the disliked games to suppress recommendations resembling them. Exclude every game in the input set from the candidate list. Filter remaining candidates to games whose stated minimum and maximum player counts include the group's size. Evaluate with Precision@10 , Recall@10 , and NDCG@10 (relevant = held-out rating of 8 or above, fixed in advance), plot accuracy against the number of games the user enters, and report how much the player-count filter (tested at 2, 3, 4, and 5 players) reduces NDCG.